SOST 20142
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Exploring the Factors Contributing to Housing Prices: The Role of School Quality in the Greater Boston Area

Word count: 1248

1.Introduction

In many urban areas, housing prices are influenced not only by property characteristics, but also by access to public services, particularly education. In the Greater Boston area, school quality varies across towns and may therefore influence residential location decisions. Black (1999) found that school district boundaries significantly affect housing prices in the United States, suggesting that households are willing to pay premiums for access to better public schools. Similarly, Simon and Polutnik (2026) found that higher 4th-grade mathematics proficiency was positively associated with residential property values in Greater Boston.

Building on this literature, this study investigates the determinants of housing prices in the Greater Boston area, with emphasis on the role of school quality after controlling for neighbourhood characteristics.

The hypotheses tested are as follows:

H_1: Higher school quality is associated with higher housing prices.

H_2: Higher household income is associated with higher housing prices.

H_3: Greater distance from Boston is associated with lower housing prices.

2.Variables and Data Preparation

2.1 Data and variables

Variable type	Variable name	Description	Measurement type	Recoding
dependent	lnHomePrice	Natural logarithm of median home price	continuous	Log transformation
independent	EDMathGrade4	Percentage of 4th-grade students proficient in math	continuous	None
control	lnHHIncome	Natural logarithm of median household income	continuous	Log transformation
control	DistBoston	Distance to Boston in miles	continuous	None
control	DistBostonSq	Distance to Boston squared (miles2)	continuous	Quadratic transformation
control	AvgClassSize	average number of students at primary school	continuous	None
control	ResTaxRate	Residential tax rate in %	continuous	None

Table2.1.1: Variables, Justifications, and Research Relevancy

The variables presented reflect the dataset after data cleaning and preprocessing procedures, resulting in 152 observations (see Code 2.1.1).

Variable	skewness
Homeprice	2.021
lnhomeprice	0.861

Table 2.1.2 Skewness before and after log transformation

Table 2.1.2 shows the logarithm of housing prices and household income was used to reduce right-skewness and improve the interpretability of the regression coefficients.

2.2 Descriptive statistics:

Variable	Mean	Median	Min	Max	Std.Dev
lnHomePrice1	13.17	13.12	12.54	14.39	0.38
EDMathGrade4	55.68	56.00	25.00	86.00	14.16
lnHHIncome	11.42	11.43	8.50	12.164	0.39
DistBoston	25.72	25.40	0.00	59.60	11.65
DistBostonSq	796.13	645.17	0.00	3552.16	621.99
AvgClassSize	18.28	18.25	13.70	22.30	1.72
ResTaxRate	14.96	14.81	6.29	22.64	2.69

Table 2.2 Descriptive statistics for variables

3. Preliminary Data analysis

3.1 Visualisation



Figure 3.1.1. Scatterplot of 4th Grade Math Proficiency Against Log Median Home Price

Figure 3.1.1 displays the bivariate relationship between 4th-grade mathematics proficiency and the natural logarithm of median housing prices.

A linear regression line with a 95% confidence band (grey area) is overlaid. The figure reveals a positive association: towns with higher percentages of students scoring proficient in mathematics tend to have higher median housing prices. The relationship appears approximately linear, with no obvious curvature or nonlinear pattern.

However, considerable dispersion around the regression line indicates that school quality alone does not fully explain variation in housing prices. This motivates the additional control variables to better isolate the effect of school quality on housing prices.



Figure 3.1.2 Boxplot of Log Housing Prices by School Quality Level

Figure 3.1.2 presents boxplots comparing log housing prices between towns with above-median ("High") and below-median ("Low") 4th-grade mathematics proficiency. The median log housing price for high-quality school towns is approximately 13.65, compared to 13.05 for low-quality towns, which is a difference of roughly 0.60 log units. This shows approximately 82% higher median prices in high-quality school towns (exp (0.60) ≈ 1.82). The low-quality group exhibits more outliers, particularly on the lower end.

While Figure 1 establishes a positive correlation, Figure 2 quantifies the approximate magnitude of this difference. Together, both visualizations support the hypothesis that better school quality is associated with higher housing prices.

3.2 Correlation Analysis

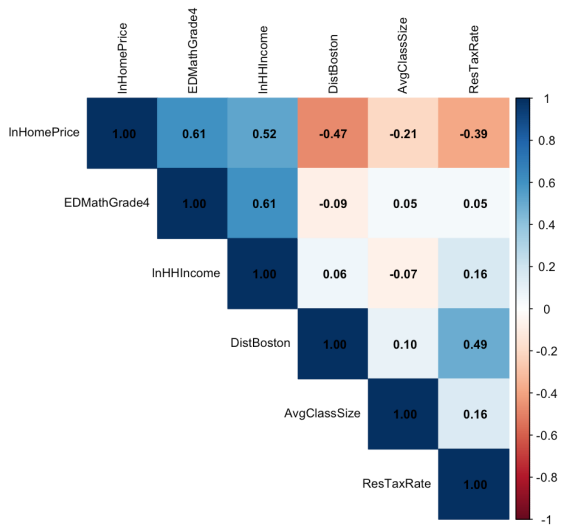


Figure 3.2 Correlation Matrix of Selected Variables

Figure 3.2 presents a correlation matrix of all continuous variables. The strongest positive correlation with lnHomePrice is EDMathGrade4 (0.61), followed by lnHHIncome (0.52). DistBoston shows a moderate negative correlation (-0.47), while AvgClassSize (-0.21) and ResTaxRate (-0.39) exhibit weaker negative correlations. DistBoston and DistBostonSq are highly correlated (0.97) because the squared term is mechanically derived from distance. Therefore, the two coefficients should be interpreted jointly to capture a potential nonlinear relationship between location and housing prices.

While the correlation matrix provides preliminary evidence for the hypothesis, this does not account for confounding factors. Therefore, multiple regression analysis is used to control for other relevant variables.

4.Regression Analysis

4.1 Modelling strategy

After data cleaning (see Code 2.1.1), the final sample contains 152 observations.

To examine the relationship between school quality and housing prices, three linear regression models were estimated. Model 1 includes only the main explanatory variable, EDMathGrade4, to examine the baseline relationship between school quality and housing prices. Model 2 adds socioeconomic and geographic control variables, including household income, distance to Boston, average class size, and residential property tax rates. Model 3 presents the full specification by additionally including the DistBostonSq to test for potential non-linear effects. The regression model is specified as follows (Model 3): $ln(HousePrice) = \beta_0 + \beta_1 EDMathGrade4 + \beta_2 ln(HHIncome) + \beta_3(DistBoston) + \beta_4(DistBostonSq) + \beta_5 AvgClassSize + \beta_6(ResTaxRate) + \epsilon$, where ϵ represents the error term.

Variable	EDMathGrade4	lnHHIncome	DistBoston	DistBostonSq	AvgClassSize	ResTaxRate
VIF	1.6752	1.6882	17.3557	16.3066	1.0592	1.4380

Table 4.1: Variance Inflation Factors (VIF)

In the multicollinearity check (Table 4), the high VIF values for DistBoston (17.09) and DistBostonSq (16.04) are expected due to the natural correlation between a variable and its square. Although this multicollinearity may inflate standard errors, DistBostonSq is retained as a cautious test for nonlinearity rather than strong evidence of a nonlinear effect.

4.2 Regression model results

Comparison of Regression Models Explaining Housing Prices

	Dependent variable:		
	ln(HomePrice)		
	Model1	Model2	Model3
	(1)	(2)	(3)
EDMathGrade4	0.0161***	0.0105***	0.0108***
	(0.0017)	(0.0015)	(0.0015)
lnHHIncome		0.3286***	0.3368***
		(0.0564)	(0.0563)
DistBoston		-0.0097***	-0.0191**
		(0.0017)	(0.0060)
DistBostonSq			0.0002
			(0.0001)
AvgClassSize		-0.0275**	-0.0285**
		(0.0101)	(0.0101)
ResTaxRate		-0.0410***	-0.0386***
		(0.0074)	(0.0075)
Constant	12.2764***	10.2041***	10.1743***
	(0.0987)	(0.6345)	(0.6314)
Observations	152	152	152
R^2	0.3696	0.7017	0.7069
Adjusted R^2	0.3654	0.6915	0.6948
Residual Std. Error	0.2991 (df = 150)	0.2085 (df = 146)	0.2074 (df = 145)
F Statistic	87.9628*** (df = 1; 150)	68.6789*** (df = 5; 146)	58.2890*** (df = 6; 145)
Note:			*p<0.05; **p<0.01; ***p<0.001

Table 4.2.1: Regression results in three models

Table 4.2.1 reports the results from the three regression specifications. Model 1 estimates the baseline bivariate relationship between school quality and housing prices. The coefficient on EDMathGrade4 is positive and statistically significant ($\beta = 0.0161$, $p < 0.001$), suggesting that towns with higher mathematics proficiency rates tend to exhibit higher housing prices.

Model 2 introduces socioeconomic and geographic controls. Although the coefficient on EDMathGrade4 decreases to 0.0105 after controls, it remains highly statistically significant ($p < 0.001$). This suggests that

the relationship between school quality and housing prices is not solely driven by broader neighbourhood characteristics.

Model 3 is the preferred specification because it includes the most comprehensive set of explanatory variables and exhibits the highest explanatory power ($R^2 = 0.7069$). This suggests that the selected predictors jointly explain a substantial proportion of the variation in housing prices. The coefficient on EDMathGrade4 remains positive and highly statistically significant ($\beta = 0.0108$, $p < 0.001$), implying that a one percentage-point increase in 4th-grade mathematics proficiency is associated with an approximately 1.08% increase in housing prices, holding other factors constant. This finding provides strong support for the main hypothesis (H_1).

Among the control variables, household income has a positive and statistically significant effect on housing prices ($\beta = 0.3368$, $p < 0.001$), supporting H_2. Distance to Boston is negatively associated with housing prices ($\beta = -0.0191$, $p < 0.01$), supporting H_3 and suggesting that properties located further from the city centre tend to be less valuable.

Average class size and residential property tax rates both exhibit negative and statistically significant relationships with housing prices. The squared distance term is not statistically significant, suggesting limited evidence of non-linearity in the relationship between distance from Boston and housing prices. Nevertheless, its inclusion slightly improves overall model fit.

Overall, the F-statistics indicate that the regression models are jointly statistically significant ($p < 0.001$). School quality remains an important and statistically robust determinant of housing prices in the Greater Boston area.

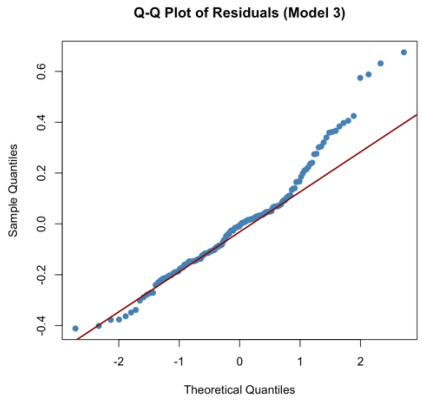


Figure 4.2.2 Q-Q Plot of Residuals (Model 3)

In Figure 4.2.2, residuals approximately follow the reference line, although slight deviations are observed in the upper tail, suggesting some mild departure from normality at higher quantiles.

5.Discussions and conclusions

This study examined the determinants of housing prices in the Greater Boston area, with emphasis on the school quality. The findings indicate that higher levels of 4th-grade mathematics proficiency are associated with significantly higher residential property values, after controlling for household income, distance to Boston, average class size, and residential tax rates. However, the observed relationship may partly reflect residential sorting, whereby wealthier households tend to concentrate in both higher-quality school districts and more expensive neighbourhoods.

In addition to school quality, household income and proximity to Boston emerge as important predictors. Higher residential tax rates and larger classroom sizes are negatively associated with housing prices, implying that fiscal conditions and educational resources may influence residential demand.

However, there are several limitations. First, the analysis relies on cross-sectional observational data and therefore cannot establish causality. Second, some relevant determinants of housing prices, such as crime rates and transportation access, are not included. Their omission may create an upward bias in the estimated effect of school quality if these factors are positively correlated with both educational performance and housing demand. Finally, the findings are limited to the Greater Boston area and may not generalize to other regions.

Overall, the results show that school quality constitutes an important correlate of housing prices in Greater Boston. The findings contribute to a broader understanding of how educational outcomes may be capitalised into residential property markets.

References

Black, S. E. (1999). *Do better schools matter? Parental valuation of elementary education. The Quarterly Journal of Economics, 114*(2), 577–599. https://doi.org/10.1162/003355399556070

Simon, J. K., & Polutnik, L. (2026). *Housing prices and school quality: A quantitative exercise using Greater Boston housing data*. *Journal of Statistics and Data Science Education*. https://doi.org/10.1080/26939169.2026.2634614

Appendix

```
rm(list = ls())
# 1.Load packages and import data
setwd("~/Desktop/study/year2/stats/assessment")
#1. Load packages
library(psych)
library(tidyverse)
library(corrplot)
library(car)

# 2.Variables and Data Preparation
data <- read_csv("Housing_price_school_quality.csv")
data$DistBostonSq <- data$DistBoston^2
my_vars <- data %>% select(lnHomePrice,
EDMathGrade4,lnHHIncome,DistBoston,DistBostonSq,AvgClassSize,ResTaxRate)
# 2.1.1Check missing values
colSums(is.na(my_vars))
nrow(my_vars)
data_clean <- na.omit(my_vars)
# 2.2.2test for skewness
skewness <- function(x) {
  x <- x[!is.na(x)]
  mean((x - mean(x))^3) / sd(x)^3
}
skewness_table <- data.frame(
  Variable = c("HomePrice", "lnHomePrice"),
  Skewness = c(skewness(data$HomePrice), skewness(data$lnHomePrice)) )
skewness_table
  # 2.2 summary
describe(data_clean)

# 3.Visualisations
# Create a categorical variable for boxplot (high/low school quality)
median_math <- median(data_clean$EDMathGrade4)
data_clean$SchoolQualityGroup <- ifelse(data_clean$EDMathGrade4 > median_math, "High",
"Low")
data_clean$SchoolQualityGroup <- factor(data_clean$SchoolQualityGroup, levels =
c("Low", "High"))
```

```r
summary(data_clean[, c("lnHomePrice", "EDMathGrade4")])
# Figure 3.1.1: Bivariate Relationship (Scatterplot)
ggplot(data_clean, aes(x = EDMathGrade4, y = lnHomePrice)) +
  geom_point(alpha = 0.6, size = 2, color = "steelblue") +
  geom_smooth(method = "lm", se = TRUE, color = "darkred") +
  labs(
    title = "Relationship Between 4th Grade Math Scores and Log Housing Prices",
    x = "4th Grade Math Proficiency (%)",
    y = "Log of Median Home Price (in $1000s)"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 12),
    axis.title = element_text(face = "bold", size = 10)
  )
# Figure 3.1.2: Boxplot - Comparing High vs Low School Quality
ggplot(data_clean, aes(x = SchoolQualityGroup, y = lnHomePrice, fill =
SchoolQualityGroup)) +
  geom_boxplot(alpha = 0.8, outlier.color = "red", outlier.size = 2) +
  geom_jitter(width = 0.2, alpha = 0.3, size = 1) +
  scale_fill_manual(values = c("Low" = "salmon", "High" = "steelblue")) +
  labs(
    title = "Housing Prices by School Quality Level",
    x = "School Quality (Above/Below Median Math Proficiency)",
    y = "Log of Median Home Price (in $1000s)"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 12),
    axis.title = element_text(face = "bold", size = 10),
    legend.position = "none"
  )
# Figure 3.2 correlation matrix
install.packages("corrplot")
cor_vars <- data_clean[, c("lnHomePrice", "EDMathGrade4", "lnHHIncome",
                     "DistBoston", "AvgClassSize", "ResTaxRate")]
cor_matrix <- cor(cor_vars, use = "complete.obs")
corrplot(cor_matrix, method = "color", type = "upper",
         addCoef.col = "black", number.cex = 0.8,
         tl.cex = 0.8, tl.col = "black",
         mar = c(0, 0, 2, 0))

# 4.Regression Analysis
# 4.1 Check missing values
colSums(is.na(my_vars))
nrow(my_vars)
data_clean <- na.omit(my_vars)
NROW(data_clean)
```

```r
# multicollinearity check
library(car)
vif(model3)

#4.2 Regression 1 (only school quality)
model1 <- lm(lnHomePrice ~ EDMathGrade4,
             data = data_clean)
summary(model1)
# Regression 2 (add more control term but without squared distance term)
model2 <- lm(lnHomePrice ~ EDMathGrade4 + lnHHIncome +
               DistBoston + AvgClassSize + ResTaxRate,
             data = data_clean)
summary(model2)
# Regression 3 (Full model with quadratic distance term)
model3 <- lm(lnHomePrice ~ EDMathGrade4 + lnHHIncome +
              DistBoston + DistBostonSq + AvgClassSize + ResTaxRate,
            data = data_clean)
summary(model3)
# table
install.packages("stargazer")
library(stargazer)
stargazer(model1, model2, model3,
          type = "html",
          title = "Comparison of Regression Models Explaining Housing Prices",
          dep.var.labels = "ln(HomePrice)",
          column.labels = c("Model1", "Model2", "Model3"),
          digits = 4,
          star.cutoffs = c(0.05, 0.01, 0.001),
          out = "regression_table.html")
# Q-Q plot
qqnorm(resid(model3),
       pch = 19,
       col = "steelblue",
       main = "Q-Q Plot of Residuals (Model 3)")

qqline(resid(model3),
       col = "darkred",
       lwd = 2)
```